I have a requirement to make an HRMS and vendor management kind of system. The situation is like this: one PSU company hires vendor employees through NICSI, and they want a way to manage purchase orders, Work orders, Advance payments, vendor employee details, and attendance.
It happens like this: Suppose there are multiple projects in the company and there is a requirement for more manpower, so the project manager can raise a purchase order request through SAP, and the purchase department will handle the purchase order process; we only need to record and track the Purchase order number(PO No.) and other basic PO details, the NICSI will

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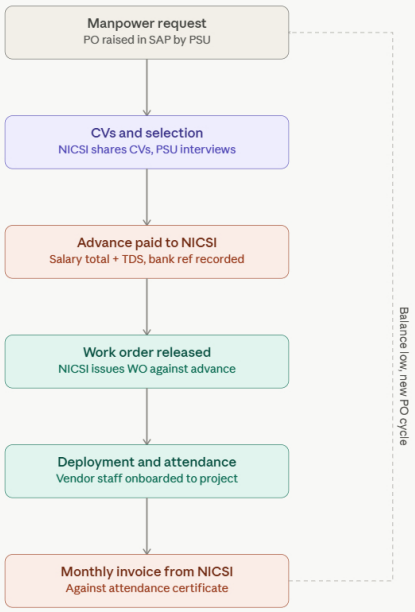
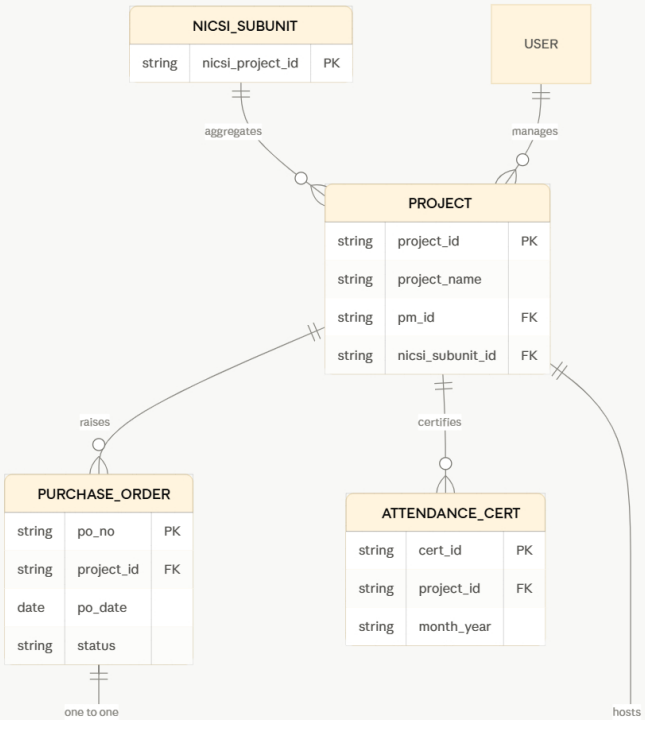
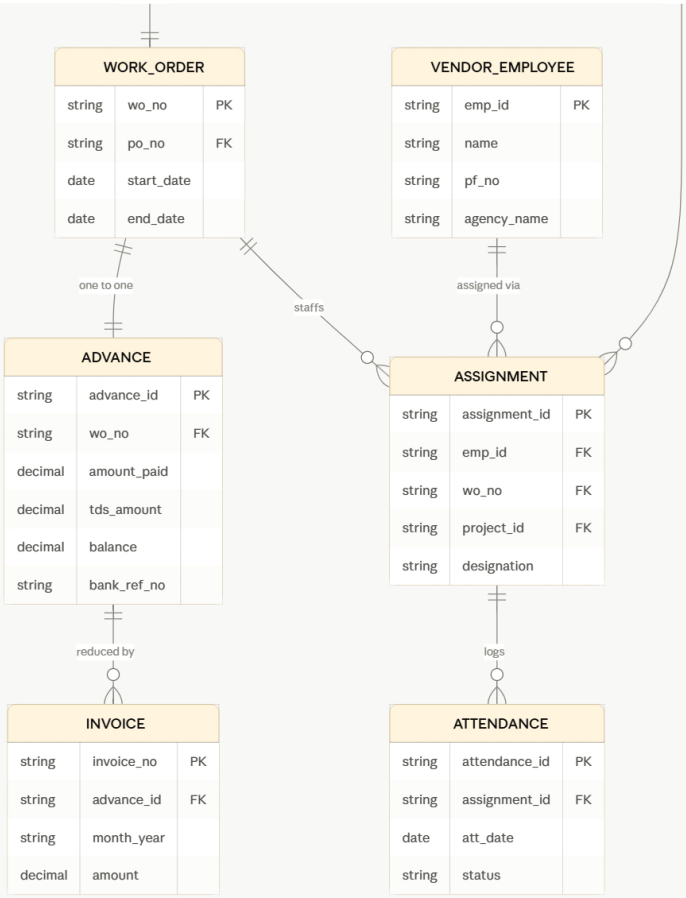
Architected comprehensive system design encompassing data models and workflows



This is the recurring lifecycle for one project: each time the advance runs low and manpower is still needed, it loops back into a fresh PO → WO → advance cycle, while everything stays tagged to the same PSU project ID.

Now let's look at how the data itself should be structured underneath this flow.





A few design decisions worth calling out before the full write-up:

The NICSI-subunit vs PSU-project split is the trickiest part of this model. NICSI only issues one project ID per PSU subunit, but you want multiple internal project IDs. So `NICSI_SUBUNIT` sits above `PROJECT` as a one-to-many parent — every PSU project rolls up to the single NICSI ID for subunit-level advance totals, while each PSU project still gets its own PO/WO/advance trail for project-level tracking.

PO ↔ WO is a strict 1:1, and Advance is 1:1 with WO — so the "advance" really lives at the WO level, and a project accumulates advance history across multiple PO→WO→Advance triples over its lifetime (your 6-month renewal example).

Employees are modeled as an assignment, not a direct link — `VENDOR_EMPLOYEE` master data is separate from `ASSIGNMENT`, because the same person can move between work orders/projects over time, and attendance needs to trace back to a specific WO/advance so consumption is attributed correctly.

Here's the full design doc with workflow states, business rules, dashboard specs, and an architecture/tech recommendation:

Created a file, read a file

The doc covers the full data model, workflow states, roles, dashboard specs, and architecture recommendations. A few things worth deciding before you start building, flagged at the end of the doc:

- Whether CV shortlisting deserves its own tracked entity or stays informal
- Whether advance entries need a maker-checker approval given how sensitive bank refs/TDS are
- What triggers a "low balance" alert (percentage vs fixed amount)
- Any data-residency requirement for a PSU (on-prem vs government cloud vs public cloud)

Happy to go deeper on any single piece next — e.g. the actual API endpoint list, or wireframes for the advance/attendance dashboards.

Nicsi vendor hrms design
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